Training Quantum Self-Attention Model in Near-Term Quantum Computer

Junyuan He
Institute of Advanced Technology
University of Science and Technology of China
Hefei, China

Yin Kan
AHU-IAI AI Joint Laboratory
Anhui University
Hefei, China

Cheng Xue
Institute of Artificial Intelligence
Hefei Comprehensive National Science Center
Hefei, China
xcheng@iai.ustc.edu.cn

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***Abstract*—Quantum computing is rapidly emerging as a revolutionary computing paradigm with significant development potential. Variational quantum algorithms, especially those employing variational quantum circuits, have demonstrated notable advantages in domains including finance, chemistry, and machine learning. However, the practical deployment of these algorithms on Noisy Intermediate-Scale Quantum (NISQ) devices faces challenges due to quantum noise, state collapse, and limited qubits numbers. The real-world performance of these algorithms remains largely unexplored. This paper addresses this gap by proposing and implementing a novel method for training quantum self-attention model for text classification directly on quantum computing chips. This initiative marks the first instance of employing such models on actual quantum hardware. Our approach optimizes quantum circuit designs to mitigate the impact of quantum noise and introduces parallel training strategies to enhance efficiency. Our experiments, conducted on the "Wukong" superconducting 72-qubit quantum computer, demonstrate that our models can be effectively trained on real quantum hardware. The results from the actual quantum chips slightly outperformed those from simulators, confirming the practical feasibility of applying quantum self-attention model to natural language processing tasks. These results validate the effectiveness of our approach and provide valuable insights for the future deployment of quantum machine learning models.**

***Index Terms*—quantum computing, quantum machine learning, near-term quantum computer**

I. Introduction

Quantum computing is widely seen as a promising new computing paradigm with enormous potential [1]. It has demonstrated significant advantages in various practical applications. With rapid advancements in quantum algorithms and hardware, quantum computing shows immense potential in fields such as finance [2], chemistry [3], optimization [4], and machine learning [5], [6].

Currently, quantum hardware is in the Noisy Intermediate-Scale Quantum (NISQ) era [7]. These devices suffer from random quantum noise, issues with quantum state collapse and decoherence, and typically consist of only a few dozen qubits [8], limiting their practical application range [9]. Despite these challenges, researchers are investigating the potential for demonstrating quantum advantage on NISQ devices. Variational Quantum Algorithms (VQAs), which leverage Variational Quantum Circuits (VQCs), have shown encouraging results in this area [10].

While many studies based on VQCs have achieved remarkable results in theory or on quantum simulators, their performance on real quantum chips remains largely unknown and challenging [11]. This paper aims to address these issues by training a quantum self-attention model on real quantum computing chips. The research proposes a method for parallel training of quantum self-attention models for text classification on real quantum chips. Additionally, we optimize quantum circuit designs to mitigate the impact of quantum noise and introduce parallel training strategies to enhance efficiency.

Our work demonstrates the feasibility of conducting natural language processing tasks directly on quantum devices, navigating the challenges posed by NISQ-era technology. The experimental outcomes, particularly with the MC and RP datasets, confirm that our quantum models can be effectively trained on real-world quantum systems. Specifically, these experiments were conducted on the "Wukong" 72-qubit superconducting quantum computer, showcasing a significant milestone in quantum computing applications.

II. Preliminaries

A. Variational Quantum Algorithms

On near-term quantum computer devices, running complete quantum algorithms typically presents considerable challenges. In contrast, VQAs provide a more feasible approach. VQAs generally consist of two main components: VQCs and classical optimization algorithms [12].

VQCs are composed of a series of trainable parameter quantum gates. By tuning these parameters, the quantum state of the circuit can be altered, affecting the algorithm's output.

In VQAs [13], the quantum circuit typically comprises three parts: Data Encoding, Ansatz Circuit, and Measurement.

a) Data Encoding: The primary function of the data encoding circuit is to map classical data into quantum states. For a classical input $x_i \in \mathbb{R}^d$, we first use a quantum encoder U_{enc} to encode the d-dimensional classical data into the n-qubits Hilbert space, i.e.,

$$|\psi_{in}\rangle = U_{enc}(x_i)|0^n\rangle. \tag{1}$$

Common encoding methods include basis encoding, angle encoding, and amplitude encoding [14].

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b) Ansatz Circuit: The ansatz circuit is essentially a parameterized quantum circuit, where the parameters θ_k are adjusted during the optimization process to extract the encoded quantum state. This step resembles the process of optimizing training parameters in classical machine learning. An ansatz can be represented as a combination of parameterized quantum gates:

$$U(\theta) = \prod_{k=1}^{m} U_k(\theta_k). \tag{2}$$

c) Measurement: The measurement part is used to measure the outcomes of the quantum circuit, often requiring multiple measurements to obtain the expectation value of a measurement operator M, which is written as:

$$\langle M \rangle = \langle \psi_{out} | \, M \, | \psi_{out} \rangle \,, \tag{3}$$

where $|\psi_{out}\rangle$ is the quantum state obtained through the Ansatz. The expectation value is then passed to subsequent classical neural networks or optimization algorithms.

Classical optimization algorithms are used to adjust the parameters θ of the parametrized quantum circuit to minimize or maximize an objective function $C(\theta)$ [15]. The objective function is typically related to the specific problem's goal, such as energy minimization or maximizing expectation values. The optimization process can be expressed as:

$$\theta^* = \arg\min_{\theta} C(\theta). \tag{4}$$

Common classical optimization algorithms include gradient descent, Newton's method, and quasi-Newton methods [16]. The update rule for gradient descent is:

$$\theta_{t+1} = \theta_t - \eta \nabla C(\theta_t), \tag{5}$$

where η is the learning rate, and θ_t and θ_{t+1} are the values of θ at time steps t and $t+1$, respectively.

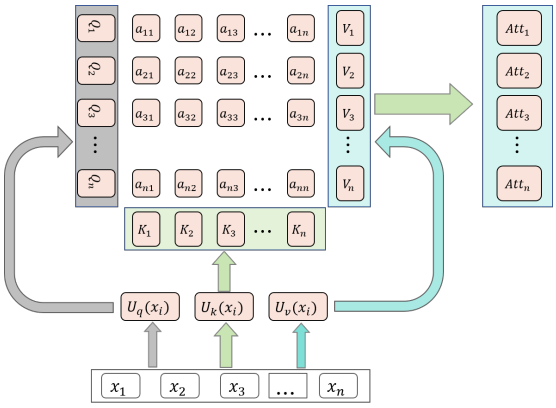


Fig. 1. Sketch of a quantum self-attention model.

B. Self-Attention Model

The self-attention model is a core component of the Transformer model [17], fundamentally altering the approach to natural language processing tasks [18]. Given input data $X \in \mathbb{R}^{n \times d}$, where n represents the number of tokens in the sequence and each token is represented by $x_i \in \mathbb{R}^d$, the self-attention model enhance the ability to deeply understand the inner dependencies between tokens.

In this model, each vector x_i in the input sequence is transformed into a query vector Q_i through the parameter matrix W_Q. Similarly, x_i is also transformed into a key vector K_i and a value vector V_i through the parameter matrices W_K and W_V, respectively. These parameter matrices W_Q, W_K, and W_V are dynamically learned during the model training process to map the input vectors into new vector spaces for subsequent attention calculations.

The mathematical expressions are as follows:

$$Q_i = W_Q \cdot x_i, \quad K_i = W_K \cdot x_i, \quad V_i = W_V \cdot x_i. \tag{6}$$

After computing these vectors, the attention scores between positions i and j are calculated using a dot product function $a_{ij} = Q_i \cdot K_j^T$. This score reveals the importance or relevance of token j with respect to token i, aiding in a deeper understanding of contextual relationships within the input sequence.

Finally, by weighted summing over all positions of the V vectors using softmax function, the attention model enhances the representation of token i, making it more enriched:

$$\text{Attention}(i) = \sum_j \text{softmax}(a_{ij}) \cdot V_j. \tag{7}$$

This process enables the model to capture complex dependencies within the input sequence, significantly enhancing its ability to generate coherent and context-aware outputs across various natural language processing tasks.

III. Parallel Quantum Self-Attention Model

A. Quantum Self-Attention Model

In the quantum self-attention model [19], the classical trainable parameter matrices W_Q, W_K, W_V are replaced with VQCs U_q, U_k, U_v. The overall process of quantum attention is shown in the Fig. 1. Specifically, for the input data $\{x_1, x_2, \ldots, x_n\}$, the process is as follows:

1) Input x_1 into the VQC $U_q(x_1)$, and run the quantum circuit and measure the expectation value to obtain Q_1:
$$Q_1 = \langle 0 | U_q^\dagger(x_1) O_q U_q(x_1) | 0 \rangle. \tag{8}$$
2) Similarly, obtain K_1 and V_1 by measuring the expectation values of the circuits $U_k(x_1)$ and $U_v(x_1)$:
$$K_1 = \langle 0 | U_k^\dagger(x_1) O_k U_k(x_1) | 0 \rangle. \tag{9}$$
$$V_1 = \langle 0 | U_v^\dagger(x_1) O_v U_v(x_1) | 0 \rangle. \tag{10}$$
3) Repeat the process for $x_2, \ldots, x_n$ to obtain Q_i, K_i, V_i for all $i \in \{1, 2, \ldots, n\}$.

After obtaining the query vector Q_i, key vector K_i, and value vector V_i for each input token x_i, the subsequent processing is the same as the classical self-attention mechanism previously described, and will not be repeated here.

Assuming that the VQCs $U_q(x_i)$, $U_k(x_i)$, and $U_v(x_i)$ have similar structures, let T_{anz} denote the time required to obtain the expectation value from running the quantum circuit once.

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For an input batch of size bs with each data point containing n tokens, the forward training time is:

$$\text{Training Time} = 3 \times bs \times n \times T_{anz}. \tag{11}$$

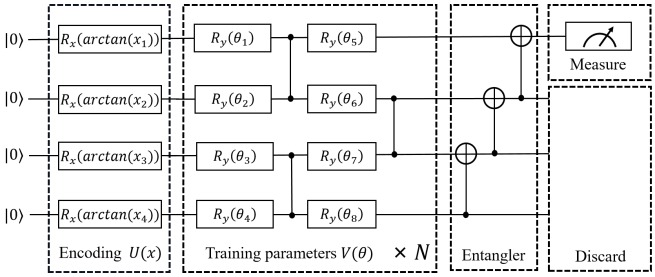


Fig. 2. Quantum circuit used in Q and K.

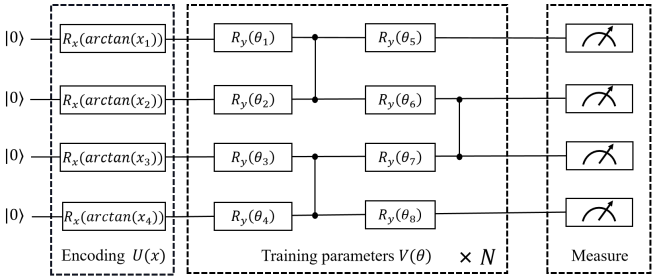


Fig. 3. Quantum circuit used in V.

B. Quantum Circuit Design

VQCs fundamentally involve rotating quantum states around the x-axis, y-axis, or z-axis on the Bloch sphere. Our circuit diagram is shown in Fig. 2 and Fig. 3. To efficiently represent classical data in quantum space, for input data $x_i \in \mathbb{R}^D$, we first compute $\hat{x_i} = \arctan(x_i)$ to compress the input data into the range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. The input data is then encoded into quantum states using the $R_x(\hat{x_i})$ gate.

For the parameterized circuit layers, we use an alternating arrangement of parameterized $R_x(\theta_i)$ gates and CZ gates, where θ_i are trainable parameters. To enhance the entanglement of the quantum system without making the circuit too complex for implementation on real quantum chips, the circuit employs alternating odd and even layers of CZ gates.

During the measurement phase, the quantum circuits $U_q(x_i)$ and $U_k(x_i)$ that generate Q_i and K_i only measure a single qubit. This simplifies the computation of the score for Q_i and K_j from a vector inner product to the multiplication of two numbers. To ensure that measuring a single qubit yields results comparable to measuring all qubits, CNOT gates are applied to all qubits before measurement to achieve sufficient entanglement.For the measurement in $U_v(x_i)$, we still measure all qubits. This maintains the dimensional consistency between the output and input of quantum self-attention, facilitating the subsequent application of residuals.

C. Parallel Strategies

Considering that VQCs can only be computed serially and not in parallel like matrix multiplications, several feasible parallel strategies are proposed below. Assuming the VQCs $U_q(x_i)$, $U_k(x_i)$, and $U_v(x_i)$ each require s qubits, three feasible parallel strategies are proposed based on the principles of the quantum self-attention model: attention-level, token-level, and batch-level parallelism.

- **Attention Level Parallelism**: For a given input token x_i, the circuits $U_q(x_i)$, $U_k(x_i)$, and $U_v(x_i)$ can be computed in parallel. This requires $3s$ qubits and improves the computation speed by a factor of 3.
- **Token Level Parallelism**: For input data $X = \{x_1, x_2, \ldots, x_n\} \in \mathbb{R}^{n \times d}$, multiple tokens x_i can be computed in parallel for their respective circuits $U_q(x_i)$, $U_k(x_i)$, and $U_v(x_i)$. For instance, if two tokens $\{x_1, x_2\}$ are computed in parallel, it requires $2s$ qubits and doubles the training speed. Theoretically, for l (where $1 \leq l \leq n$) parallel processes, ls qubits are required, increasing the computation speed by $\left\lceil \frac{n}{l} \right\rceil$.
- **Batch Level Parallelism**: For a neural network with batch size $= bs$, assuming input data Input $= \{X_1, X_2, \ldots, X_{bs}\}$ with $X_i \in \mathbb{R}^{n \times d}$, multiple data points X_i can be computed in parallel. Similarly, for k (where $1 \leq k \leq bs$) parallel processes, ks qubits are required, increasing the computation speed by $\left\lceil \frac{bs}{k} \right\rceil$.

By combining the above three parallel strategies, the highest parallel efficiency can be achieved by performing attention-level parallelism for $U_q(x_i)$, $U_k(x_i)$, and $U_v(x_i)$ simultaneously, token-level parallelism for n parallel processes, and batch-level parallelism for bs parallel processes. This approach increases the speed by:

$$3 \times bs \times n \tag{12}$$

times compared to the non-parallel strategy. For example, in a model with an average sentence length of 20 tokens and a batch size of 16, this strategy can speed up the training process by 960 times.

D. Parameter Optimization Strategy

The Parameter-Shift Rule (PSR) is a method used in quantum computing to evaluate the gradient of a quantum circuit with respect to its parameters [20]. It is particularly useful for optimizing parameterized quantum circuits in VQAs. The key idea behind PSR is to shift the parameters of the quantum gate by a fixed amount and use the resulting measurements to estimate the gradient.

Given a parameterized quantum circuit with a parameter θ, the gradient of an observable O with respect to θ is given by:

$$\frac{\partial \langle O \rangle}{\partial \theta} = \frac{1}{2}\left(\langle O(\theta + \frac{\pi}{2})\rangle - \langle O(\theta - \frac{\pi}{2})\rangle\right), \tag{13}$$

where $\langle O(\theta)\rangle$ is the expectation value of the observable O when the parameter is θ. The parameters are shifted by $\frac{\pi}{2}$ in the positive and negative directions to compute the gradient.

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This method avoids finite-difference approximations and provides precise computation of gradients for quantum circuits [21].

IV. Experiment in Near-Term Quantum Computer

A. Experiment Setup

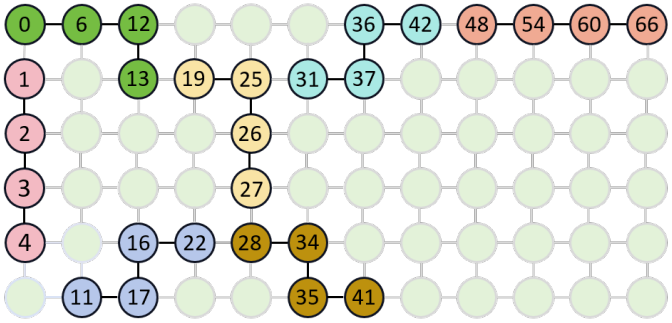


Fig. 4. The topology diagram of quantum chip.

TABLE I
INDEX OF QUBITS USED IN EACH GROUP

Group	q0	q1	q2	q3
Group 1	0	6	12	13
Group 2	1	2	3	4
Group 3	22	16	17	11
Group 4	19	25	26	27
Group 5	28	34	35	41
Group 6	42	36	37	31
Group 7	48	54	60	66

Our experiments were conducted on the "Wukong" superconducting 72-qubit quantum computer manufactured by OriginQ. The quantum computer has an average single-qubit gate fidelity of 0.9969, an average CZ gate fidelity of 0.9637, an average T_1 time of 14.78 μs, and an average T_2 time of 1.87 μs.

We built our quantum self-attention model using the VQNet quantum machine learning framework, which facilitates easy access to quantum computers for training our model. Our computational tasks were submitted via the quantum cloud platform.

We selected a seven-way parallel scheme, using 28 qubits, to conduct our experiments. The topology diagram of the quantum real chip is shown in the Fig. 4 and Tab. I, where the groups with darker colors are the ones we use. Since we cannot directly extract the quantum state, multiple samples of the quantum circuit are needed to obtain the output results of the quantum circuit. In our experiments, we ran each quantum circuit 2000 times. The actual runtime for a single circuit on the quantum computer is 100 μs, so after multiple measurements, each quantum circuit requires 0.2 s.

B. Experiment Dataset

We selected two publicly available binary text datasets in the textual domain, namely MC dataset and RP dataset [22]. The MC ('meaning classification') dataset consists of 130 sentences, with 70 used for training, 30 for validation, and 30 for testing. Each sentence contains 3 or 4 words. Half of these sentences describe food-related topics, while the other half describe information technology (IT). Each topic consists of 65 sentences, such as:

```
"skillful programmer creates software"
"chef prepares delicious meal"
```

The task in MC is to classify each sentence into either the food or IT category. There are a total of 17 different words in MC, and the vocabulary is shared between both categories. Therefore, this task is non-trivial. The other dataset is a more conceptually oriented dataset called RP ("relative pronoun"). The RP dataset selects 105 noun phrases containing relative clauses from the RelPron dataset. The selection criterion ensures each word appears at least 3 times in the dataset, resulting in a total vocabulary of 115 words. RP is also a binary classification problem, aiming to predict whether a given noun phrase contains a subject relative clause ("device that detects planets") or an object relative clause ("device that observatory has"). The motivation behind this task is its requirement for some grammatical sensitivity from the model, making it a reasonable choice for testing quantum self-attention models.

C. Result

In theoretical analysis, without parallel strategies, the time required for forward propagation in each step of the training process can be calculated as follows:

$$\begin{aligned} \text{forwardTime} &= \text{bs} \times \text{seqLen} \times \text{circuitNum} \times \text{time} \\ &= 7 \times 4 \times 3 \times 0.2 \\ &= 84 \times 0.2 \\ &= 16.8 \text{ seconds}, \end{aligned} \tag{14}$$

where bs represents the batch size, seqLen is the number of tokens per sentence in the batch, time indicates the running time of each circuit, and circuitNum denotes the number of quantum circuits. In our experiment, we have three circuits, U_q, U_k, and U_v.

During the backward propagation phase, The estimated total runtime is:

$$\begin{aligned} \text{backwardTime} &= \text{forwardTime} \times \text{paramsNum} \times \text{PSR} \\ &= 16.8 \times 8 \times 2 \\ &= 16.8 \times 16 \\ &= 268.8 \text{ seconds}, \end{aligned} \tag{15}$$

where forwardTime represents the estimated time of forward, paramsNum represents the number of parameters and PSR is the number of parameter shift rules.

In practical training scenarios, the time for forward propagation is approximately 85 seconds, and backward propagation takes about 270 seconds. The discrepancy between theoretical and actual values mainly arises from network communication delays and machine queuing times. The experimental training process curves conducted on two datasets are shown in Fig. 5

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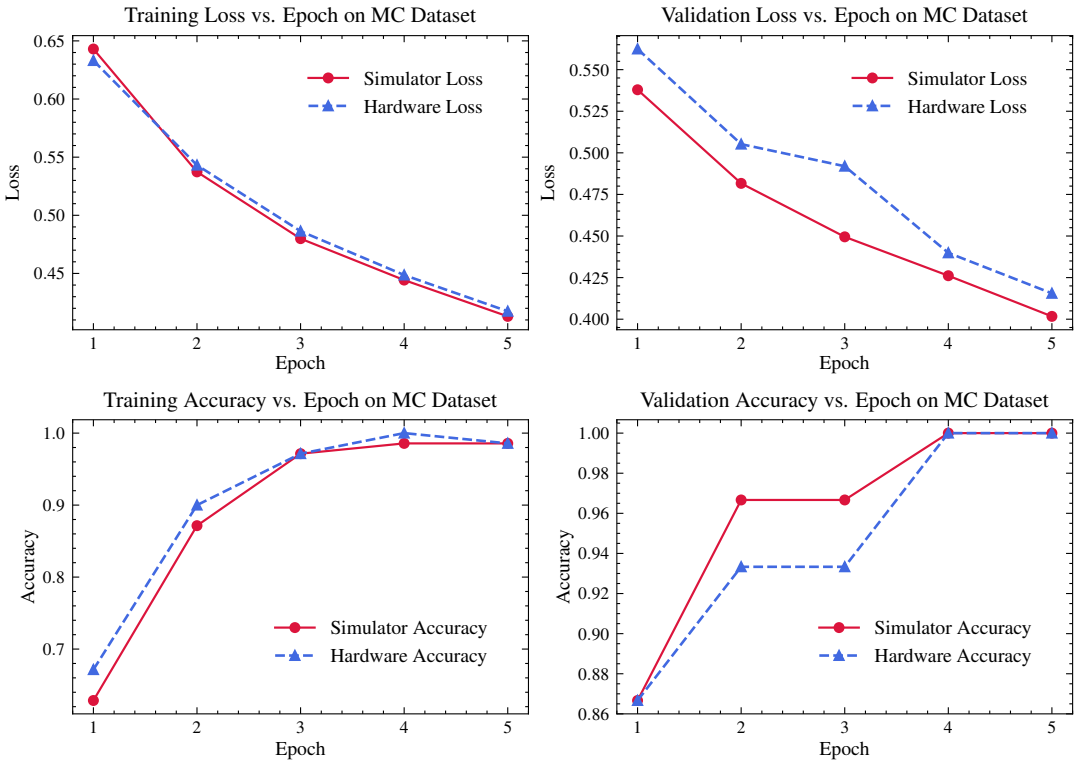


Fig. 5. Training and test loss and accuracy over epochs on MC dataset.

TABLE II
EXPERIMENT RESULT ON SIMULATOR AND HARDWARE

Platform	Acc of MC	Acc of RP	Forward Time	Backward Time
hardware	100.00%	83.87%	85s	270s
simulator	100.00%	80.66%	5567s	90921s

and Fig. 6, respectively. Tab. II shows the accuracy of the simulator and hardware on the MC dataset and RP dataset test sets, as well as the forward and backward propagation time for each step. On the MC dataset, although the accuracy of both the real machine and the simulator can reach 100% on the test set, the real machine has higher accuracy on the training set and lower training and testing set losses when converging. On the RP dataset, both the training and testing sets have lower loss values and higher accuracy compared to the simulator.

V. Conclusions

This research has demonstrated the practical training of quantum self-attention model on actual quantum computing hardware, specifically using the "Wukong" 72-qubit superconducting quantum computer. Our work demonstrates that it is feasible to conduct natural language processing tasks directly on quantum devices, navigating the challenges posed by NISQ-era technology. The experimental outcomes, particularly with the MC and RP datasets, confirm that our quantum models can be effectively trained on real-world quantum systems, showcasing a significant milestone in quantum computing applications. Future efforts will aim to refine these methods, enhance the scalability of quantum models, and expand their applicability across diverse machine learning challenges. This endeavor sets a foundation for advancing quantum computing into more complex domains and potentially achieving quantum advantage in computational efficiency.

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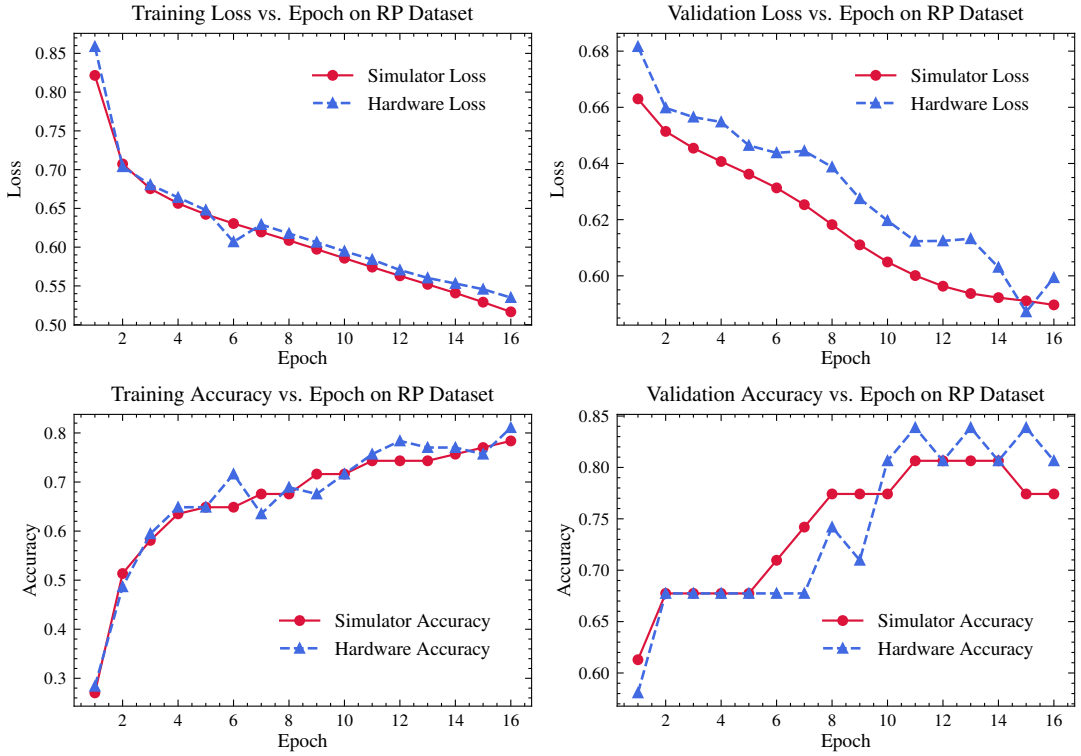


Fig. 6. Training and test Loss and accuracy over epochs on RP dataset.

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